

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First Semester of M.Tech. (CE/IT/EE/EC) Examination****March- Apr 2018****CE767 Operating System Design & Concepts****Date: 05.04.2018, Thursday****Time: 10:00 a.m. To 1:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q – 1 (a)** Explain the process state transition diagram for multiprogramming environment. **[03]****(b)** Under what circumstances process moves from user mode to kernel mode? **[04]****Q - 2 (a)** Explain the scenarios for retrieval of a buffer in buffer cache. **[05]****(b)** Explain in detail about PCB and also explain how process represented in system memory. **[04]****(c)** List major categories of Operating System services. Explain any of them in detail. **[04]****OR****(c)** What resources are used when a thread is created? What resources are used when a process is created? **[04]****Q - 3** **Answer Following Questions (Any Three)** **[15]****(a)** Explain an algorithm for open () system call.**(b)** When pid=fork () is called, what are the sequence of operation executed by kernel?**(c)** Describe the various scenarios that can happen in algorithm breada.**(d)** Explain the architecture of Google file system.**(e)** Compare and contrast the handling of system calls and of exceptions. What are the similarities and differences?**SECTION – II****Q - 4 (a)** Draw and explain in brief the architecture of UNIX system. **[03]****(b)** How does UNIX kernel maintain the consistency of the system data structures? **[04]****Q - 5 (a)** Explain the scenarios when new inode is assigned a disk inode to a newly created file. **[05]****(b)** List and explain any two major sources of signals. **[04]****OR**

- (a) Differentiate followings: [05]
- (i) Synchronous and Asynchronous process execution
 - (ii) Delayed Write and Asynchronous write
- (b) Which are basic method(s) to protect processes from each other? Explain any of them in detail. [04]
- (c) Explain unlink race condition. [04]

Q - 6 Answer Following Questions (Any Three) [15]

- (a) Explain an algorithm for namei which converts pathname to inode.
- (b) What is shared memory? Explain how to make the shared memory segment available using proper diagram.
- (c) Explain the algorithm for mknod system call or creation of special files.
- (d) Explain System V IPC Mechanism using Semaphore.
- (e) Write down an algorithm for mounting a File system.
